

CALCULUS & DIFFERENTIAL EQUATIONS

Examination Year: 2026 | Maximum Marks: 100 | Total Questions: 15 | Time Allowed: 3 Hours

GENERAL INSTRUCTIONS:

- 1. All questions are compulsory unless specified otherwise.
- 2. Candidates must show all working, intermediate steps, and derivations to earn full marks.
- 3. Draw diagrams wherever necessary; label all parts clearly.
- 4. Figures, graphs and diagrams are provided with the respective question.
- 5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total
Marks	6	7	8	7	8	6	7	8	7	7	6	7	8	7	8	100

Question 1 [6 Marks]

Find the general solution to the first-order linear differential equation: $(dy / dx) + 2xy = x e^{-x^2}$.

Question 2 [7 Marks]

Compute the Jacobian matrix for the vector-valued function $f(x,y,z) = (x^2 y, \sin(yz), e^{x+z})$.

Question 3**[8 Marks]**

Evaluate the line integral $\int_C (x^2 + y^2) \, ds$ where C is the semicircle $x^2 + y^2 = 4$ from $(2,0)$ to $(-2,0)$.

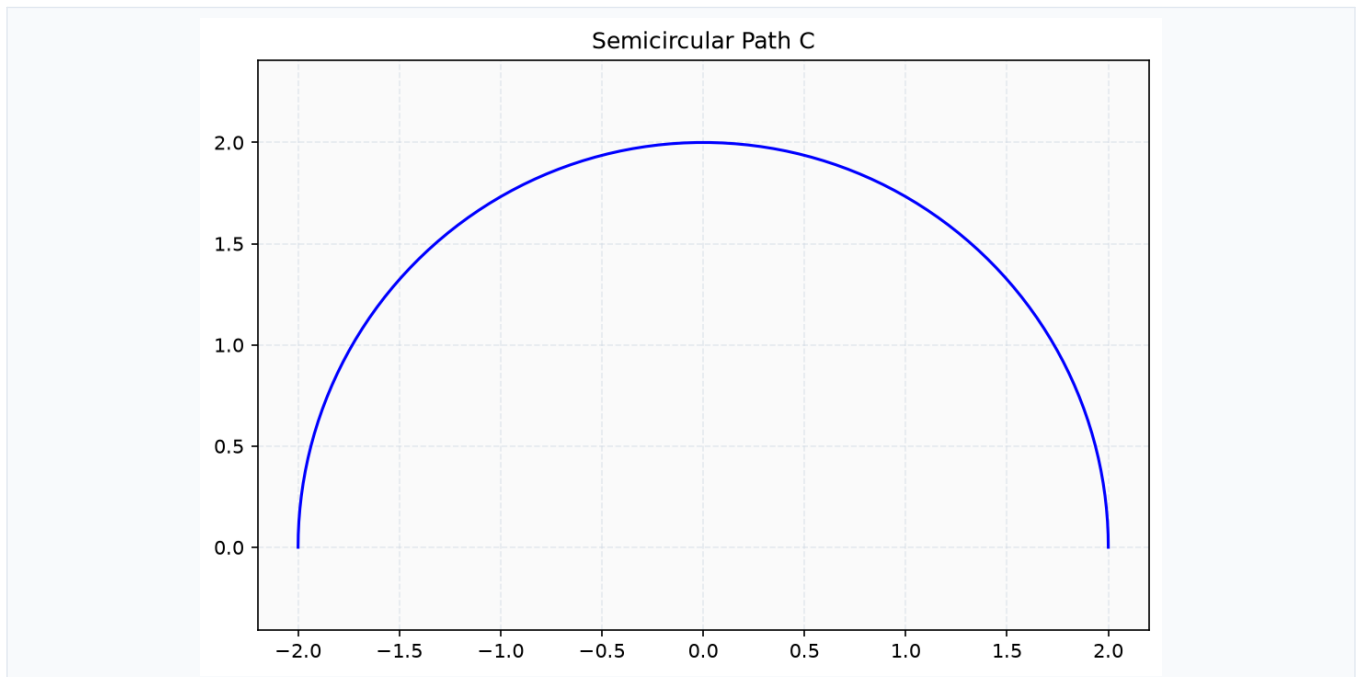


Figure 1

Question 4**[7 Marks]**

Solve the exact differential equation: $(2xy + y^3)dx + (x^2 + 3xy^2)dy = 0$.

Question 5**[8 Marks]**

Find the critical points of $f(x,y) = x^3 + y^3 - 3xy$ and classify them as local maxima, minima or saddle points.

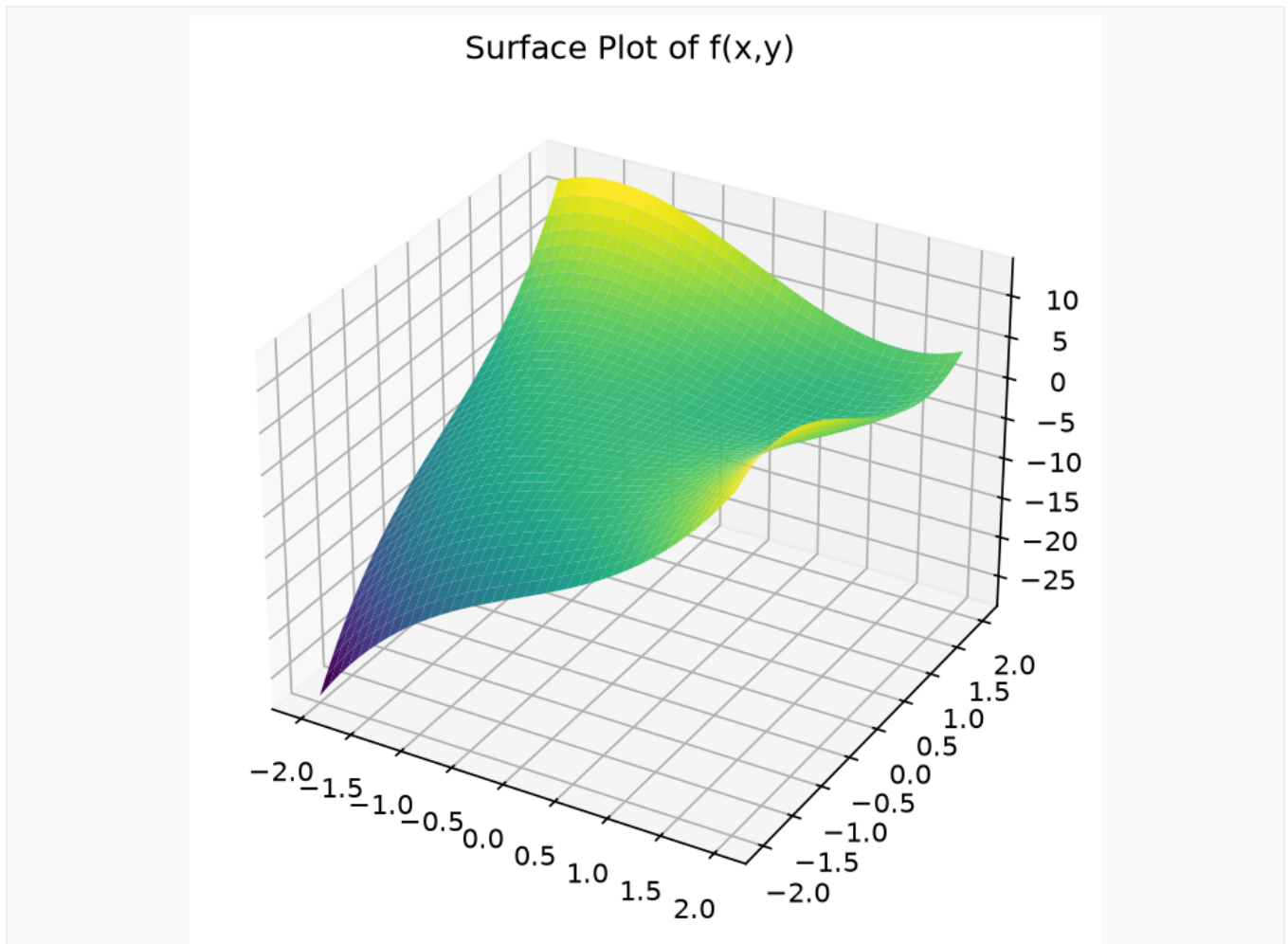


Figure 2

Question 6**[6 Marks]**

Determine the radius of convergence for the power series: $\sum_{n=1}^{\infty} ((2x-3)^n / n 5^n)$.

Question 7**[7 Marks]**

Solve the Bernoulli equation: $(dy / dx) + (y / x) = x^2 y^3$.

Question 8**[8 Marks]**

Compute the flux of $\mathbf{F} = (x^2, y^2, z^2)$ across the surface of the unit cube $[0,1]^3$.

Question 9**[7 Marks]**

Find the general solution to the second-order ODE: $y'' - 4y' + 4y = e^{2x} \cos x$.

Question 10**[7 Marks]**

Evaluate the triple integral $\int_E z \, dV$ where E is bounded by the planes $x=0$, $y=0$, $z=0$ and $x+y+z=1$.

Question 11**[6 Marks]**

Solve the initial value problem: $y' = (x^2 / y(1+x^3))$ with $y(0) = 2$.

Question 12**[7 Marks]**

Find the Fourier series representation of $f(x) = x^2$ on the interval $[-\pi, \pi]$.

Question 13**[8 Marks]**

Use Green's Theorem to evaluate $\oint_C (y^2 \, dx + x^2 \, dy)$ where C is the boundary of the region between $y = x^2$ and $y = x$.

Question 14**[7 Marks]**

Solve the system of differential equations: $(dx / dt) = 2x - y$, $(dy / dt) = x + 2y$ with initial conditions $x(0) = 1$, $y(0) = 0$.

Question 15**[8 Marks]**

Find the Laplace transform of $f(t) = t^2 \sin(3t)$ and use it to solve $y'' + 4y = t^2 \sin(3t)$ with $y(0) = y'(0) = 0$.
